

Portfolio Project: BLDC Remote Control Fan PCB

PCB Reverse Engineering & Design Modernization

Project Details

Project Type:	Consumer Electronics PCB Reverse Engineering & Circuit Improvement
Product Category:	BLDC Motor Fan Controller (Remote Controlled)
Scope of Work:	PCB tracing, component identification and rating analysis, hand-drawn schematic reconstruction, circuit verification through simulation, design improvement, KiCad schematic and PCB layout, BOM and manufacturing file generation
Tools / Technologies:	KiCad, Proteus, Multimeter, AI-Assisted Engineering Analysis
Role:	Lead Electronics Engineer

1. Project Background

This project involved reverse engineering a commercially available **BLDC remote-controlled fan controller PCB**, originally manufactured as a closed-source product with no schematic, no BOM, and no design documentation.

The PCB was treated as a complete black box. The objective was to **fully reconstruct the design, verify its behavior, improve weak points, and deliver a production-ready hardware design**.

This was not a copy-paste task — it required deep observation, structured thinking, and repeated verification.

2. Reverse Engineering Strategy

The work was intentionally approached in structured stages to avoid design assumptions:

- Manual PCB tracing to understand copper routing and connectivity
- Documentation of every component, package, and electrical rating
- Creation of a hand-drawn schematic to understand functional blocks
- Logical separation of power, control, and motor-drive sections

This early-stage discipline reduced later design errors.

3. Circuit Verification Using Proteus

Before committing to a CAD design, the reconstructed circuit was tested in **Proteus**.

This phase introduced several challenges:

- Many original components were unavailable in Proteus libraries
- Functional alternatives had to be selected for testing
- Simulation math and timing mismatches caused incorrect results
- Multiple retesting cycles were required to isolate real faults from simulation artifacts

Through critical thinking, manual calculations, and repeated validation, the circuit behavior was successfully verified.

4. Circuit Improvements & Enhancements

After verification, several improvements were intentionally introduced:

- Improved signal referencing for stable control behavior
- Protection enhancements to improve reliability
- Optimization of component ratings for long-term operation
- Minor logic corrections discovered during simulation

These improvements enhanced robustness while maintaining original functionality.

5. KiCad Schematic & PCB Design

Once the circuit was validated:

- The schematic was rebuilt cleanly in KiCad
- Multiple footprint mismatches were identified and resolved
- PCB routing was completed with careful attention to power paths
- Design-for-manufacturing considerations were applied

Despite footprint and library challenges, the design was completed through iterative correction and verification.

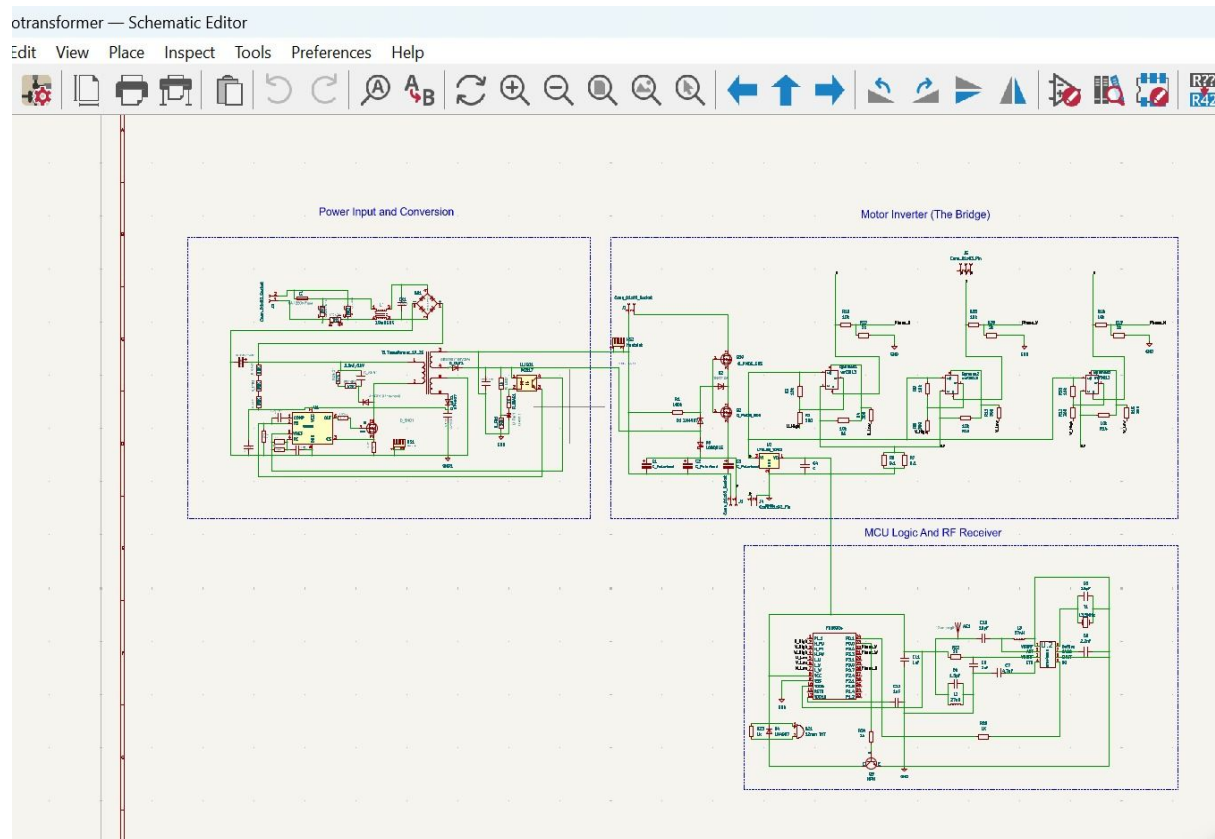
6. Manufacturing Outputs

The final deliverables included:

- Verified KiCad schematic
- Production-ready PCB layout
- Complete BOM with correct component ratings
- Pick-and-place files
- Fabrication-ready Gerber files

7. Visual Documentation

Reverse Engineered Schematic



uptotransformer — PCB Editor

File Edit View Place Route Inspector Tools Preferences Help

Track: use netclass width Via: use netclass sizes F.Cu (PgUp) 0.0100 mm (0.39 mils) Zoom 2.20

8. Summary

This project demonstrates the ability to:

- Reverse engineer undocumented consumer electronics hardware
- Validate designs using simulation under imperfect tool conditions
- Apply critical thinking when tools fail or data is missing
- Deliver a complete, manufacturable PCB design

A complex black-box system transformed into a fully understood and manufacturable design through patience, analysis, and persistence — Alhamdulillah.